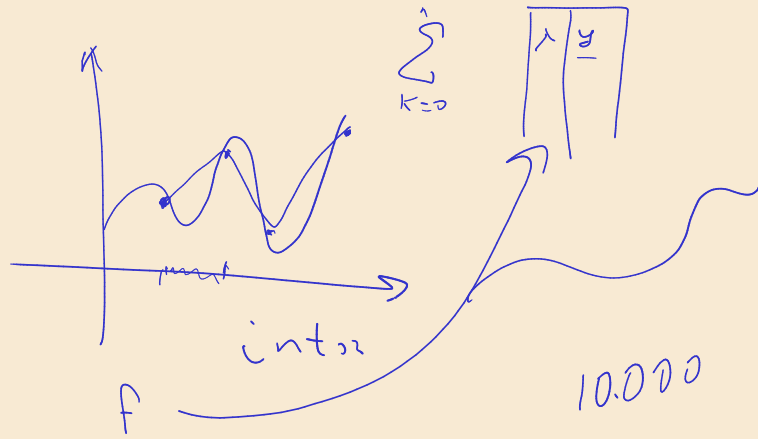
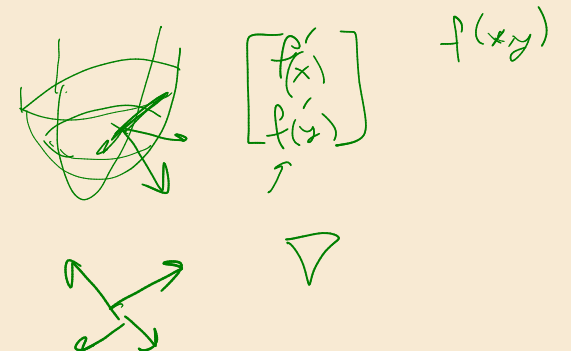
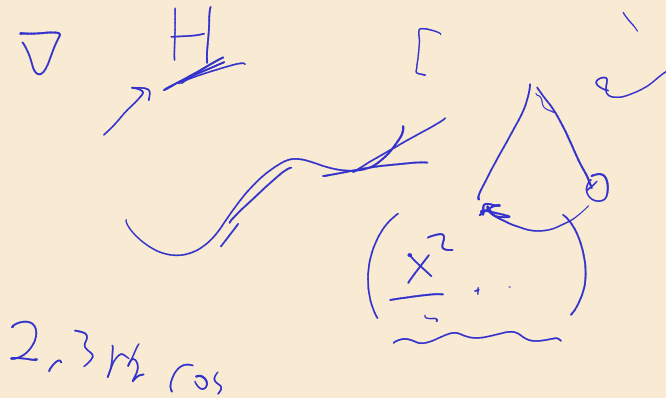


$$f(x) = \sum_{k=0}^n \underbrace{f^{(k)}(\xi)}_{k \cdot + = \cdot} \cdot \underbrace{\frac{(x-a)^k}{k!}}_{\text{Taylor polynomial}} + R_n(x)$$

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} \cdot \underbrace{(x-a)^{n+1}}_{\text{Taylor remainder}}$$



$$\begin{bmatrix} 0.4 \\ 0.8 \\ 1 \end{bmatrix}$$

②

$$D_u f(x) = \nabla f(x) \cdot u$$

$$\|u\| = 1 \quad |ab| < |a||b|$$

4th rule

$$|\nabla f(x) \cdot u| \leq \|\nabla f(x)\| \cdot \|u\|$$

$$|\nabla f(x) \cdot u| \leq \|\nabla f(x)\|$$

$$\nabla f(x)^T \cdot \nabla f(x)$$

$$u = \nabla f(x)$$

$$\begin{bmatrix} 3 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 18$$

$$3 \cdot 3 + 3 \cdot 3$$

$$V^T V$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1^2 + 2^2 = 5$$

$$\begin{bmatrix} a \\ b \end{bmatrix}^T \begin{bmatrix} a \\ b \end{bmatrix} = a^2 + b^2$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

$$\cos \alpha$$

$$\frac{1}{\sqrt{5}}$$

$$\frac{g}{z} t^2$$

$$g t$$

$$t$$

$$\frac{1}{\sqrt{2}}$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 5$$

$$A_{m,n}, G, D$$

$$0.01 \sigma$$

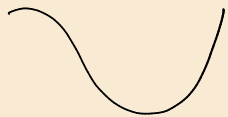
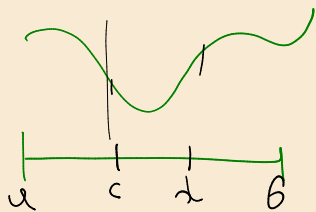
$$10$$

$$5 \cdot 2.5$$

$$2.5$$

$$\frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}}$$



5/1-4
gölmö!

$$\begin{bmatrix} a-x & b \\ c & d(x) \end{bmatrix}$$

$$Av = \lambda v$$

$$[A - \lambda] v = [0]$$

[]

$$\det(\cdot) = \lambda$$

$$\begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix}$$

$$Av = \lambda v$$

A

$$\begin{bmatrix} 3-\lambda & 0 \\ 0 & 5-\lambda \end{bmatrix} \rightarrow (3-\lambda)(5-\lambda) - 0 = 0$$

$$\lambda = 3, \lambda = 5$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

0.

